



Debargha Chowdhury

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Professional Summary

Technical Content Writer and Data Analyst with 3+ years of experience in producing structured technical documentation, analyzing public datasets, and delivering data-driven insights. Experienced in working with JSON datasets, Python scripting, notebooks, and reproducible analytical workflows. Strong ability to translate complex technical findings into clear documentation, structured narratives, and actionable business recommendations.

Education

Diploma : Electronics And Telecommunication Engineering 2013-2016

Central Calcutta Politechnic : Kolkata CGPA: 7.7

B.Tech: Information Technology 2016-2019

Government College of Engineering And Textile Technology : Kolkata CGPA: 7.59

Languages Known

English ★★★★★

Hindi ★★★★★

Bengali ★★★★★

Projects

Credit Card Fraud Detection

- Developed machine learning models using Random Forest and Logistic Regression, achieving 92% fraud-detection accuracy.
- Performed feature engineering, data validation, and class-imbalance handling to improve model performance.
- Created business-focused recommendations for fraud-risk reduction based on analytical findings and model outputs.

Interactive Data Dashboard using Streamlit

- Designed and deployed a Streamlit-based dashboard integrating classification and regression models for real-time analytical insights.
- Automated data visualization and reporting workflows, improving accessibility of model outputs for end users.
- Documented evaluation metrics, model comparisons, and deployment processes to support maintainability.

Technical Skills

- Programming & Data Tools:** Python, SQL, JSON, Pandas, NumPy, Jupyter Notebook, Scikit-learn, TensorFlow, LaTeX
- Machine Learning & AI:** Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP), Pattern Recognition, AI Model Integration, Performance Analysis, Random Forest, Logistic Regression, Decision Trees, ARIMA
- AI Coding Tools:** ChatGPT, Copilot, Gemini, Claud
- Documentation Tools:** Technical Documentation, SOP Writing, Implementation Guides
- Data Visualization:** Power BI, Tableau, Streamlit, Plotly, Matplotlib
- Testing & Optimization:** Debugging, Testing, Data Validation, Data Cleaning
- Professional Skills:** Analytical Skills, Problem-Solving Skills, Communication Skills, Team-Oriented Mindset

Professional Experience

Technical Writer

2025-Currently

SDSNGN Technology Pvt. Ltd. (On-site at UCO Bank – NTT Data MCNI Project) | Kolkata | West Bengal

- Transformed the documentation process for UCO Bank by shifting deliverables from repeated rejections by the client to a 100% approval rate for the documents I created, improving stakeholder confidence and client satisfaction.
- Developed and maintained 25+ SOPs, implementation guides, and operational documents for **CheckMK**, **ITSM**, and **NCM** platforms, strengthening process standardization and knowledge transfer across teams.
- Collaborated with engineers, system administrators, and project stakeholders to capture accurate technical workflows and align all documentation with UCO Bank requirements and review expectations.
- Introduced structured templates, version-control practices, and review checkpoints, reducing documentation inconsistencies and improving document quality, clarity, and acceptance rates.
- Created clear, traceable technical documentation that supported onboarding, operational continuity, and incident-resolution processes for multiple teams within the project.
- Assisted in data validation, reporting, and documentation quality assurance activities using structured datasets and technical workflows to support accurate project delivery.

Executive Technical Academic Researcher

2021 - 2023

Royal Research | Kolkata | West Bengal

- Produced 100+ analytical reports and research documents for students studying abroad, helping them submit high-quality academic work and achieve strong marks.
- Analysed structured and semi-structured datasets for 100+ academic projects, translating complex findings into clear, submission-ready research documents.
- Improved research reproducibility by documenting methodologies, assumptions, data sources, and analytical workflows for every assignment.
- Worked extensively with JSON and tabular datasets, performing data cleaning, transformation, validation, and exploratory analysis to support accurate reporting.
- Communicated technical findings to non-technical stakeholders through structured reports, presentations, and visual summaries, contributing to positive student feedback and satisfaction.

Mathematics Faculty and Content Creator

Cheenta Academy | Kolkata | West Bengal

2020 - 2021

- Developed educational content covering advanced mathematical and analytical concepts for students preparing for competitive examinations.
- Created structured learning materials that simplified complex topics and improved student comprehension.
- Evaluated assignments and provided detailed analytical feedback, helping students strengthen problem-solving and logical reasoning skills.

Role-based Desktop Library Management System in Python

- Built a role-based desktop Library Management System in Python, as measured by seamless coordination across 6 modular components (**Login, Register, Menu, Student, Librarian, Database**), by architecting a **Tkinter GUI** layered over a normalized 3-table SQLite backend.
- Implemented secure, role-based access control for Students and Librarians, as measured by a two-tier authentication flow that gates dashboard access behind a librarian-approval status, by writing credential-validation logic against **SQLite** user records with a pending/approved account state.

Distracted Driver Detection using Deep Learning (EfficientNetB3)

- Built a deep learning classifier to detect 10 categories of distracted driving behavior (texting, calling, drinking, reaching behind, etc.), achieving 99.58% validation accuracy (near-perfect 1.00 macro precision/recall/F1) on a 4,481-image held-out set, by fine-tuning a transfer-learning CNN (EfficientNetB3 pretrained on ImageNet) with a custom classification head.
- Improved model accuracy from 88.7% to 99.58% across training, by engineering a two-phase training pipeline — initial training with a frozen backbone, followed by fine-tuning the last 100 layers at a reduced learning rate (1e-5) — combined with early stopping and learning-rate scheduling.
- Reduced trainable parameters by ~92% (923K of 11.7M total), as measured by model summary parameter counts, by leveraging a frozen pretrained backbone with GlobalAveragePooling, batch normalization, and dense layers, cutting compute cost without sacrificing accuracy.
- Addressed class imbalance and overfitting risk on a 22,424-image dataset (10 behavior classes), as measured by consistently high per-class F1 scores (0.98–1.00), by applying computed class weights, targeted data augmentation (rotation, zoom, shift, brightness — no horizontal flip, to preserve left/right distinction), and dropout regularization.

Technical Certification

Coding Ninjas Data Analytics Job Bootcamp Program January 2026

- Completed comprehensive bootcamp covering SQL for Data Analysis
- Programming in Python
- Statistics & EDA with Python
- Data Visualization with Power BI

Deloitte Australia Data Analytics Job Simulation on Forage September 2025

- Completed job simulation involving data analysis and forensic technology techniques
- Created interactive data dashboard using Tableau for business intelligence and data-driven insights
- Utilised Excel for data classification and drew actionable business conclusions from complex datasets

Deloitte Australia Technology Job Simulation on Forage September 2025

- Completed a technology-focused job simulation involving software development and Python coding
- Wrote a comprehensive technical proposal for dashboard creation and implementation

Anthropic Academy

AI Fluency: Framework & Foundations July 2026

- Completed the AI Fluency: Framework & Foundations certification, mastering the 4Ds framework (Delegation, Description, Discernment, Diligence)
- Developed critical evaluation skills for assessing AI-generated outputs and identifying appropriate use cases
- Learned structured approaches to effective AI delegation and task-specific prompt design
- Strengthened diligence practices for verifying AI outputs, ensuring accuracy and accountability in professional contexts

Claude 101 May 2026

- Completed Anthropic's Claude 101 certification, gaining foundational knowledge of large language model capabilities and use cases
- Developed practical skills in prompt construction, context management, and effective AI-assisted communication
- Learned core principles of responsible AI usage, including model limitations and best practices for reliable outputs
- Applied AI fluency concepts to improve documentation workflows and content generation efficiency

Claude Code 101 July 2026

- Completed Anthropic's Claude Code 101 certification, building foundational skills in AI-assisted software development workflows
- Gained hands-on experience using Claude Code for code generation, debugging, and repository-level tasks
- Learned agentic coding concepts, including task delegation and iterative code review with AI tools
- Strengthened technical understanding of integrating AI coding assistants into version-controlled development environments (Git/GitHub)

Claude Platform 101 July 2026

- Completed Anthropic's Claude Platform 101 certification, covering API-based access and integration of Claude models
- Gained working knowledge of platform architecture, model selection, and deployment considerations for enterprise use cases
- Learned principles of scalable AI integration, including tool use and structured outputs
- Built foundational understanding of API-driven workflows applicable to enterprise IT and documentation tooling